

Intergenerational Transfer of White Matter Integrity in Arithmetic and Reading Networks

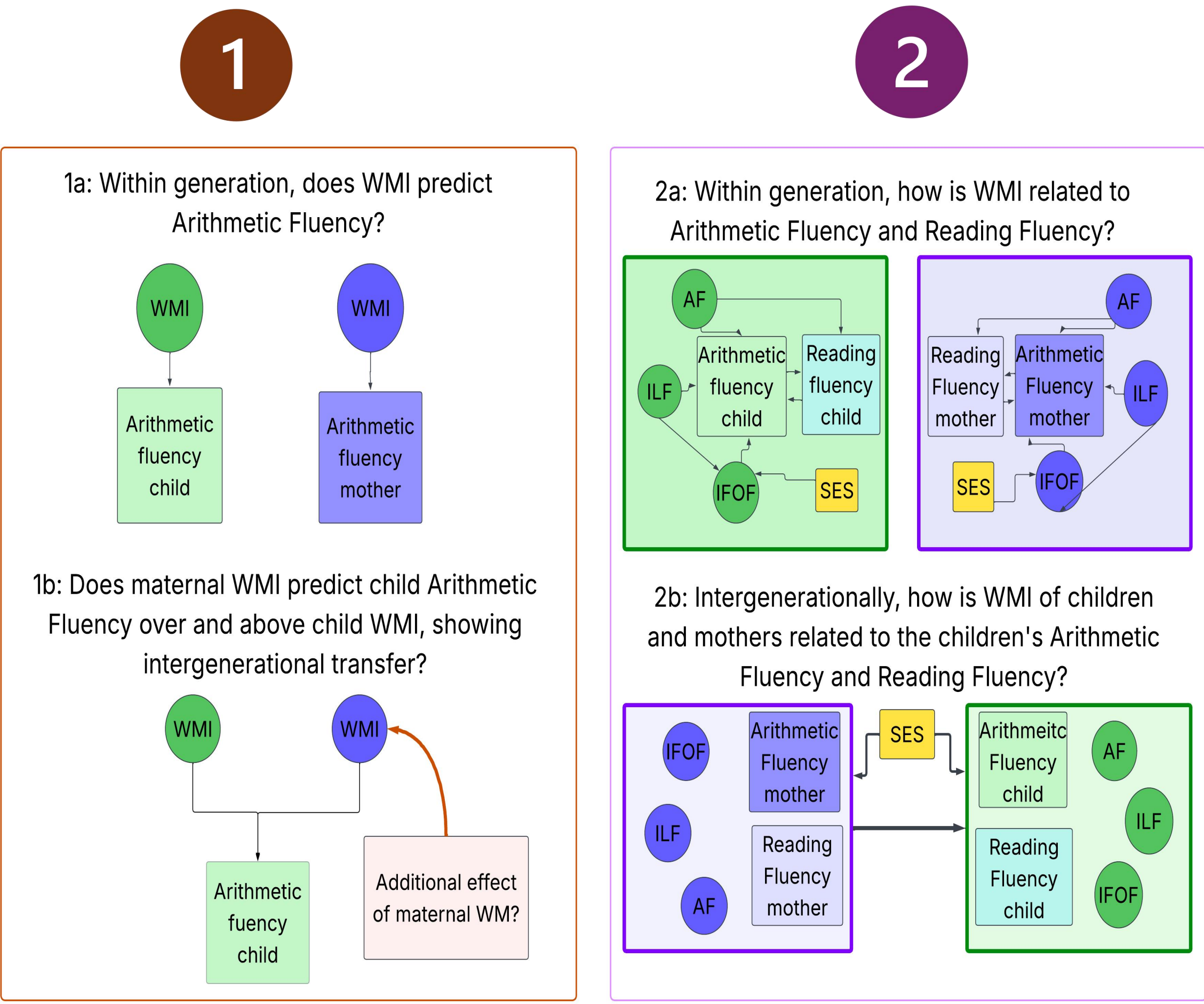
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Background

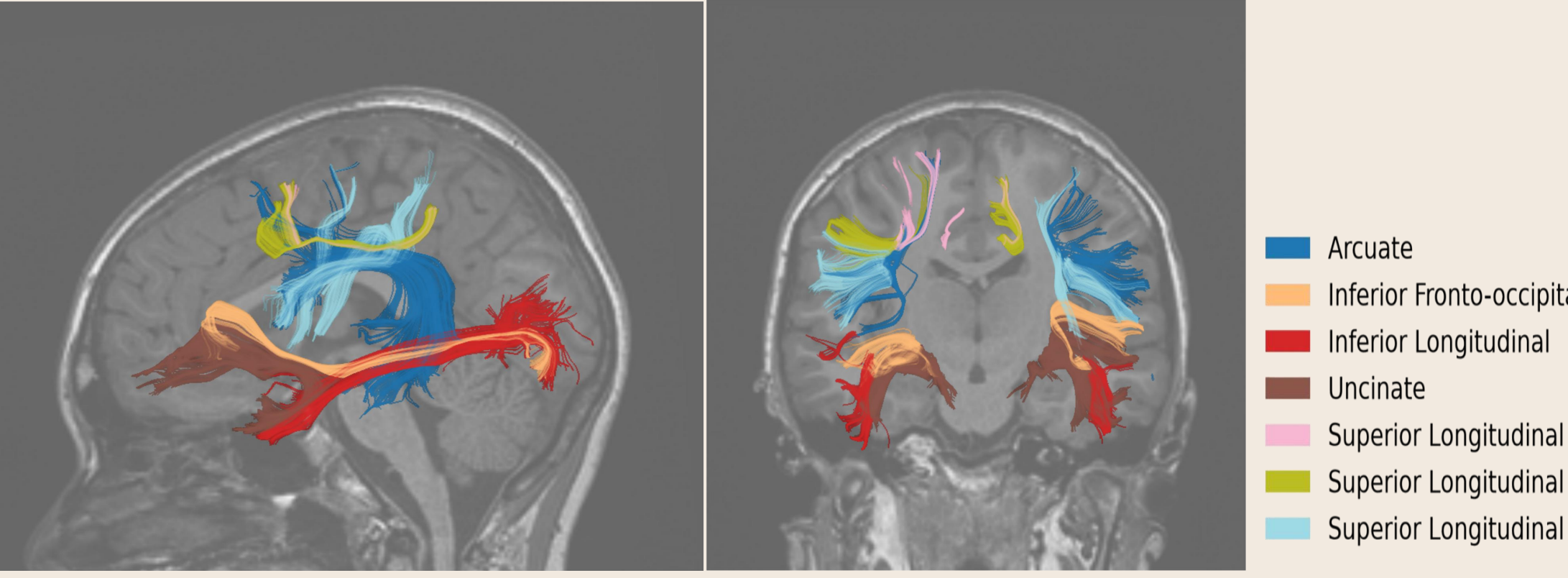
- Reading and arithmetic develop together in childhood ^{1,2}
- Arithmetic fact retrieval draws on the same phonological codes used in word decoding, required in reading ³
- Reading and arithmetic are familial skills: parent-child similarity shows up behaviourally ⁴ and neurally ⁵
- Reading and arithmetic share many brain networks, including white matter (WM) tracts like tracts linked to arithmetic (Arcuate Fasciculus—AF, Superior Longitudinal Fasciculus—SLF, Inferior Fronto-Occipital Fasciculus—IFOF, Inferior Longitudinal Fasciculus—ILF, Uncinate Fasciculus—UF) ⁶ are also linked to reading (AF, IFOF, ILF) ^{7,8}
- Intergenerational studies could offer insights into the co-development of academic skills
- Maternal IFOF integrity predicts child IFOF integrity and accounts for most of the parent-child overlap in reading development ⁹
- The correlation between arithmetic and white matter integrity (WMI) in AF disappears once reading is statistically controlled for, which prompts for studying these skills together and intergenerationally ¹⁰

Research Questions



Methods

- Participants**
- 32 mothers ($M_{age} = 44$ years)
 - 40 10-to-14-year-old children ($M_{age} = 12.38$ years, $F=11$)
 - 31 mother-child dyads after excluding siblings in a family
- Materials**
- Behavioural metrics: Reading Fluency ¹¹ and Arithmetic Fluency ¹²
 - SES: Mother's education level
- Preprocessing and Tractography**
- Preprocessing: QSIprep v 0.26.0 ¹³
 - Analysis: Anatomically constrained tractography using a deterministic pipeline of PyAFQ package ¹⁴
 - Metrics of WMI: Fractional Anisotropy (FA) and Mean Diffusivity (MD)
 - Tracts with a participant-level streamline count below 10 were excluded



1

1a: Does children's/mother's WMI (FA and MD) in the selected tracts correlate with children's arithmetic performance, controlling for SES?

Analysis	Results
Hierarchical regression, run separately for children and mothers :	
M1: Control model (SES, age, sex — only for children, and head motion)	Children: In left SLF III , the variance in arithmetic fluency was significantly (uncorrected) explained by children's MD in that tract; this did not survive FDR across 14 tracts
M2: M1 + WMI measure of the child or mother (FA/MD)	
F-test compares the two models, per tract and per metric. FDR-corrected across the 14 tracts.	Mothers: In right AF , the variance in arithmetic fluency was significantly (uncorrected) explained by mothers' MD in that tracts, but these did not survive FDR across 14 tracts .

1b: Does mothers' WMI (FA and MD) in the selected tracts predict children's arithmetic performance, over and above children's own white matter integrity?

Analysis	Results
Hierarchical regression:	
M3: M2 of children + Mother's WMI metric (FA/MD) + Mother's age	Intergenerational: In right SLF-I , mother's FA was a significant predictor over and above children's FA, surviving FDR correction .

2

2a: Within generation (children, mothers), how are FA and MD in tracts related to reading and arithmetic (AF, IFOF, ILF) associated with their Arithmetic Fluency, Reading Fluency and SES?

Analysis	Results
Gaussian graphical model (EBICglasso with $\gamma = 0.2$) Nodes: tract FA/MD & fluency residualized on age, sex & motion; SES entered raw One network per generation: children (9 nodes, N=25), mothers (9 nodes, N=23) Bootstrapped (bootnet) for edge & centrality stability	

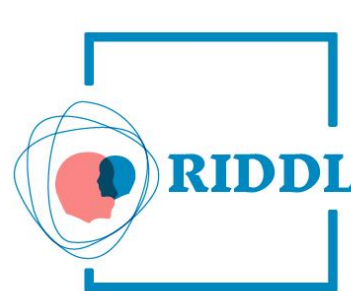
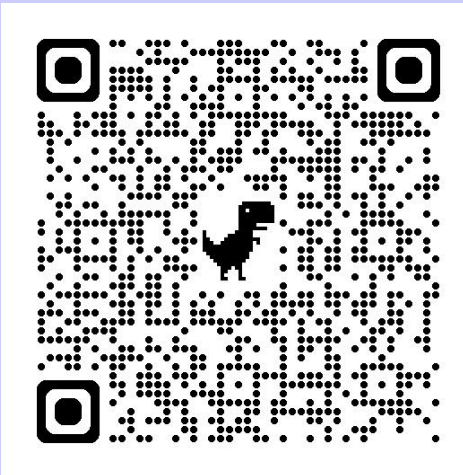
2b: Does maternal WMI in shared tracts between arithmetic and reading relate to children's WMI, reading fluency or arithmetic fluency across generations, beyond within-generation tract-behaviour associations?

Analysis	Results
Same tracts and GGM approach, now in one combined child-mother network 17 nodes, N = 20 complete dyads Cross-generation edges highlighted in the network figure Bootstrapped (bootnet) for edge & centrality stability	

Take Home!

- **MD reveals structure that FA does not:** true in every network, in both studies.
- **Right SLF-I: mother's FA predicts child's arithmetic fluency** over and above the child's FA.
- Clearest intergenerational MD link: **child L-IFOF** and **mother L-ILF**.
- Reading fluency (in children's network) and arithmetic fluency (in intergenerational network) **connect to MD of tracts**.
- **Mothers' own reading fluency stays isolated** in every network and mother's arithmetic fluency connects with the MD of children's tracts in the intergenerational network.

Poster online!



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